

Hamiltonian Dynamics and Three-body Problem

Dongho Lee

October 14th, 2025

SNU, QSMS

Presented in Rookie's Pitch, SNU



Download slides

Introduction

The **three-body problem**, came from the celestial mechanics, describes the motion of an object under the gravitational force of two bodies.

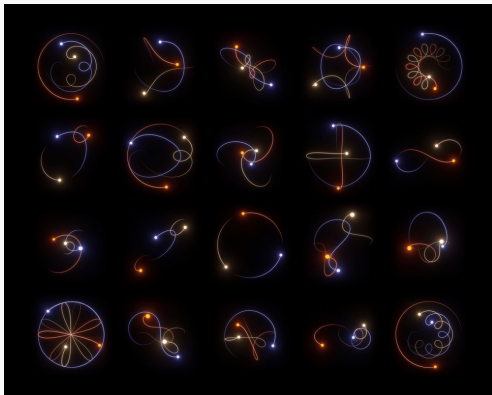


Figure 1: Various solutions of three-body problem.¹

¹https://www.wikiwand.com/en/articles/Three-body_problem

Main interest : Periodic orbits of three-body problem.

- Hamiltonian dynamics
- Periodic orbits and Bifurcation
- Three-body Problem

Hamiltonian Dynamics

Classical Mechanics

Classical mechanics is governed by 2nd order ODE on $\mathbb{R}^n$ which is given by Newton's second law,

$$F = ma \quad \Leftrightarrow \quad -\nabla V(q) = \ddot{q},$$

where $q \in \mathbb{R}^n$ is the position and V is the **potential energy**.

The **mechanical energy** is defined by

$$H = K + V := \frac{|\dot{q}|^2}{2} + V(q),$$

where K is the **kinetic energy**.

Law (Classical Energy Conserve)

H is conserved along the trajectory of q determined by force F of the form $-\nabla V$ where $V = V(q)$.

Question 1. Can we re-formulate the equation of motion via the law of energy conservation?

Consider H as a function $H = H(q, p)$ where $p \in T_q^* \mathbb{R}^n$, so

$$H = \frac{|p|^2}{2} + V(q).$$

Then we have

$$\begin{aligned}\dot{q} &= p = \frac{\partial H}{\partial p}, \\ \dot{p} &= \ddot{q} = -\nabla V = -\frac{\partial H}{\partial q}.\end{aligned}$$

Now we consider 1st order ODE on $T^* \mathbb{R}^n$.

Hamiltonian Mechanics

Let $H : T^*\mathbb{R}^n \rightarrow \mathbb{R}$ be a function, which will be called **Hamiltonian**.

Consider the vector field X_H on $T^*\mathbb{R}^n$,

$$X_H = \begin{pmatrix} \partial_p H \\ -\partial_q H \end{pmatrix} = \sum_i \frac{\partial H}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial}{\partial p_i}.$$

Theorem (Energy Conservation)

Along the trajectory of X_H , H is conserved.

We call X_H the **Hamiltonian vector field**.

Now we can impose *any* function H as the total energy of the system, and use the viewpoint of the classical mechanics.

Question 2. How can we make X_H from H ?

We have

$$dH = \sum_i \frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial p_i} dp_i,$$

so if we can change (dq, dp) to $(-\partial_p, \partial_q)$, we get X_H .

The duality via $\omega = dp \wedge dq$ works by

$$\omega(X_H, \cdot) = -dH(\cdot),$$

since

$$\omega(X_H, \partial_{q_i}) = \omega((- \partial_{q_i} H) \partial_{p_i}, \partial_{q_i}) = -\partial_{p_i} H = -dH(\partial_{q_i}),$$

$$\omega(X_H, \partial_{p_i}) = \omega((\partial_{p_i} H) \partial_{q_i}, \partial_{p_i}) = -\partial_{p_i} H = -dH(\partial_{p_i}).$$

Symplectic Structure

For suitable 2-tensor ω , we must have

1. **Non-degeneracy** ($\ker \omega = 0$) for duality.
2. **Alternating property** ($\omega(X, X) = 0$) for conservation of energy,

$$X_H(H) = dH(X_H) = -\omega(X_H, X_H) = 0.$$

3. **Closedness** ($d\omega = 0$) for the conservation of ω ,

$$\mathcal{L}_{X_H}\omega = di_{X_H}\omega + i_{X_H}d\omega = -d(dH) + i_{X_H}d\omega = i_{X_H}d\omega = 0.$$

We call closed non-degenerate 2-form ω a **symplectic form**.

A manifold W with a symplectic form ω is called **symplectic manifold**.

The study of (W, ω) is symplectic geometry / topology / dynamics.

Nice introductory texts : [Arn89], [CdS01].

Bifurcation and Index

Toy Example : Function

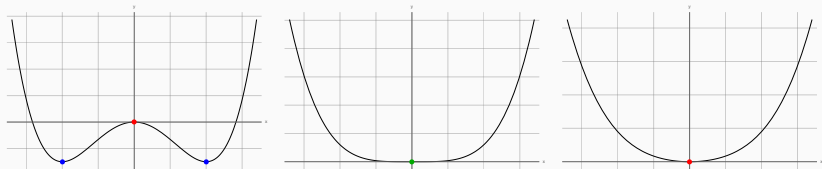


Figure 2: Toy model of a bifurcation and CZ-index

Consider the family of functions $f_t(x) = x^4 - tx^2$.

As t varies, a family of critical points changes when it degenerates.

Conley-Zehnder Index

Main interest : **Periodic orbits** of a given Hamiltonian,

$\gamma : [0, \tau] \rightarrow W$ such that $\dot{\gamma}(t) = X_H(\gamma(t))$, $\gamma(0) = \gamma(\tau)$.

A periodic orbit γ is **non-degenerate** if there is no *multiplicity*.

(Precise definition : $\det(dFl_\tau^{X_H}|_{H^{-1}(c)}(q) - \text{Id}) = \langle X_H \rangle$.)

Generically, every periodic orbit of X_H on $H^{-1}(c)$ is non-degenerate.

For non-degenerate γ , we define **Conley-Zehnder index** $\mu_{CZ}(\gamma) \in \mathbb{Z}$.

- Can be regarded as a 'symplectic rotation number'.
- Gives grading of Floer homology, which is a symplectic invariant.
 $\Rightarrow$ The indices of *all* periodic orbits are topologically controlled.
- Tools: local Floer homology, Morse-Bott spectral sequence, etc.

Bifurcation of Periodic Orbits

Let γ_c be a periodic orbit of X_H in $H^{-1}(c)$.

1. If γ_c is non-degenerate, there is an *orbit cylinder*

$$\Gamma : S^1 \times (c - \varepsilon, c + \varepsilon) \rightarrow W$$

such that $\Gamma(\cdot, c') = \gamma_{c'}(\cdot)$ is a periodic orbit of X_H in $H^{-1}(c')$.

2. If γ_c is non-degenerate for $c_0 \leq c \leq c_1$, the CZ-index of γ_c for $c_0 \leq c \leq c_1$ is constant.
3. If γ_c becomes degenerate, the index might change and new families of orbit might born, or the family γ_c might vanish.
We call this **bifurcation**.
4. Floer homology is invariant of the choice of c unless $c \in \text{Crit}(H)$, so we can track the bifurcation by computing Conley-Zehnder indices.

Bifurcation of Periodic Orbits

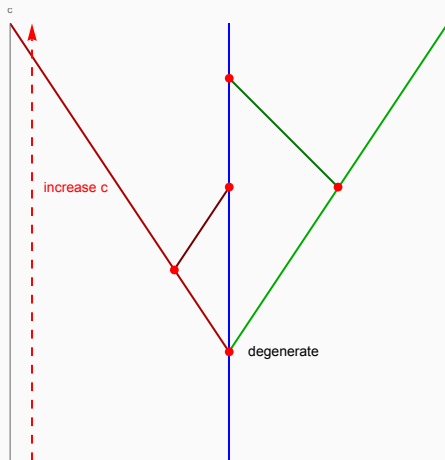


Figure 3: Bifurcation diagram

- Each point of the diagram is a periodic orbit.
- At each segment between red points, orbits are non-degenerate and the indices do not change.
- At red points, the orbit degenerates and the index changes, and families of orbits are born or vanish. (bifurcation)

Remark. Instead of changing the energy, we can perturb the Hamiltonian and see the bifurcation behavior.

⇒ Investigate a complicated system as a perturbation of simpler one.

Nice introductory texts : [Mil63], [Sma67], [AM78], [AD14]

Three-body Problem

Kepler Problem

The **Kepler problem** describes the motion of an object under the gravitational force of the other body.

$$E(q, p) = \frac{|p|^2}{2} - \frac{1}{|q|}.$$

In 17th century, Johannes Kepler established three laws from pure observation...

Theorem (Kepler's Laws)

1. *The E -orbits are conic sections with one focus at the origin.
If $E < 0$, the orbits are ellipses.*
2. *The areal velocity $d\text{Area} = r^2\dot{\theta}$ is constant along the orbit.*
3. *The period is given by $\tau = 2\pi/(-2E)^{3/2}$.*

Let two bodies E (earth) and M (moon) has mass ratio $1 - \mu : \mu$.

$$H_t(q, p) = \frac{|p|^2}{2} - \frac{\mu}{|q - M(t)|} - \frac{1 - \mu}{|q - E(t)|}$$

Assumptions

1. The third body is much lighter than M and E , so the motion of M and E are not affected by the third body.
 $\Rightarrow$ The motions of M and E are governed by Kepler problem.
2. M and E are in *circular* motion.

Circular Restricted Three-body Problem

The **circular restricted three-body problem** (CRTBP) is determined by the same $H_t(q, p)$ with

$$M(t) = (1 - \mu)(\cos t, -\sin t, 0), \quad E(t) = -\mu(\cos t, -\sin t, 0).$$

We can remove the time-dependency by using *rotating frame*, which gives

$$H(q, p) = \frac{|p|^2}{2} - \frac{\mu}{|q - M|} - \frac{1 - \mu}{|q - E|} + (p_1 q_2 - p_2 q_1).$$

This time-independent Hamiltonian also defines CRTBP.

Circular Restricted Three-body Problem

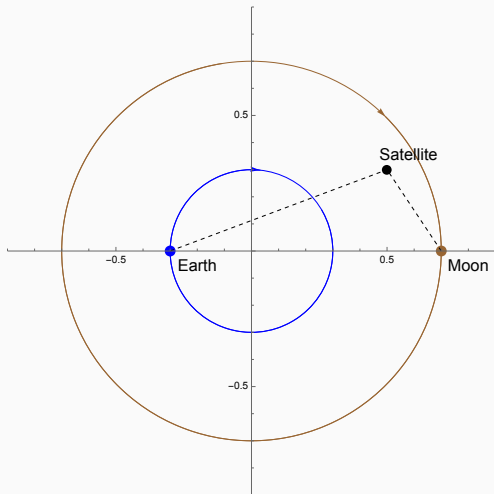


Figure 4: Illustration of CRTBP.

Hill's Region

We can rewrite the Hamiltonian as

$$\begin{aligned} H(q, p) &= \frac{1}{2} \left((p_1 - q_2)^2 + (p_2 + q_1)^2 + q_3^2 \right) - \frac{\mu}{|q - M|} - \frac{1 - \mu}{|q - E|} - \frac{q_1^2 + q_2^2}{2} \\ &= \frac{1}{2} |\tilde{p}|^2 + U(q). \end{aligned}$$

For the energy level c , we have $H(q, p) = c \Rightarrow U(q) \leq c$.

We call $\mathfrak{R}_c = \{q \in \mathbb{R}^3 : U(q) \leq c\}$ **Hill's region**.

Idea. For given energy level c , the position q must stay in $\mathfrak{R}_c$.

Lagrange Points

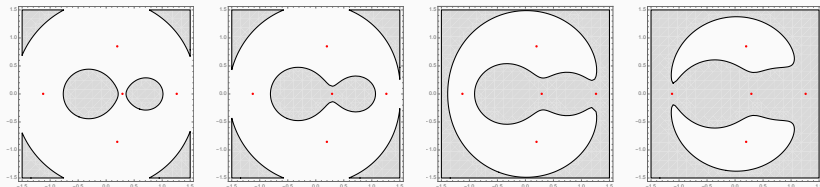


Figure 5: Hill's regions for increasing energies

There are 5 critical points (red points), called **Lagrange points**.

The topology of the Hill's region changes through $H(\ell_i)$'s.

Rotating Kepler Problem

Main interest : Periodic orbits of CRTBP.

CRTBP is still hard... $\Rightarrow$ Use limit cases to get ideas.

Rotating Kepler problem (RKP) is the limit case $\mu = 0$:

$$H(q, p) = \frac{|p|^2}{2} - \frac{1}{|q|} + (p_1 q_2 - p_2 q_1).$$

Note. The trajectory can be decomposed into the Kepler trajectory and the rotation along q_3 -axis.

$\Rightarrow$ Kepler orbits invariant under the rotation are periodic orbits of RKP.

Periodic Orbits of RKP

There are 4 non-degenerate periodic orbits of RKP for generic energy.

- Planar circular orbits : **retrograde orbit** γ_+ and **direct orbit** γ_- .
- **Vertical collision orbits** : north and south $\gamma_{c\pm}$.

Theorem ([Lee25], ArXiv preprint)

Classification of periodic orbits of RKP and computation of indices.

Periodic Orbits of RKP

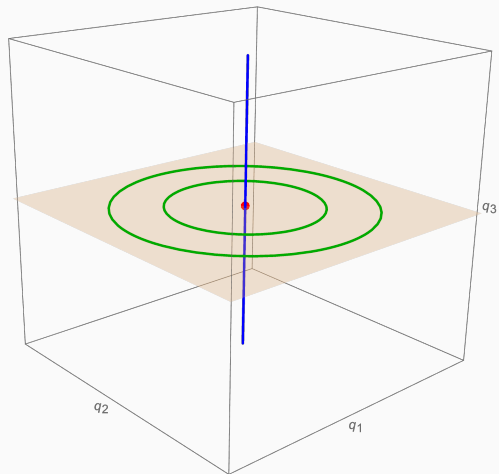


Figure 6: Non-degenerate orbits of RKP.

Retrograde Orbit of CRTBP

- We can consider CRTBP as a perturbation of RKP, and consider the family of non-degenerate orbits.
- 4 non-degenerate orbits of RKP also exist in CRTBP, which can be shown numerically, but not theoretically.

Some candidates of the definition of the retrograde orbit :

- The orbit with the smallest period (or shortest length).
- The orbit with Conley-Zehnder index 2.
- The orbit bifurcated from the retrograde orbit of RKP.
- The orbit obtained by *Birkhoff shooting method*.

Birkhoff Shooting Method

Theorem (Birkhoff)

For $0 < \mu < 1$ and $c < H(\ell_1)$, there exists a solution of CRTBP $(q_1, q_2) : [0, \tau] \rightarrow \mathbb{R} \times (-\infty, 0]$ such that

1. $q_2(0) = q_2(\tau) = 0$.
2. $q_1'(0) = q_1'(\tau) = 0$.
3. $\ell_3 < q_1(0) < -\mu < \ell_1$.

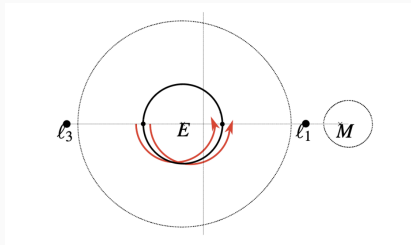


Figure 7: Birkhoff Shooting Method (courtesy of Otto van Koert)

Final Remarks

- Many numerical results about CRTBP are known, but without the aid of numerics, only little is known.
- Some simpler limit cases of CRTBP (RKP, Euler problem, Hill's lunar problem) can be studied.
- Still, CRTBP presents many interesting questions, both mathematically and practically.

Nice introductory texts : [FvK18], [Cel10]

Thank you for your attention!



References i



Mohammed Abouzaid, *Symplectic cohomology and Viterbo's theorem*, Free loop spaces in geometry and topology, IRMA Lect. Math. Theor. Phys., vol. 24, Eur. Math. Soc., Zürich, 2015, pp. 271–485. MR 3444367



Michèle Audin and Mihai Damian, *Morse theory and Floer homology*, Universitext, Springer, London; EDP Sciences, Les Ulis, 2014, Translated from the 2010 French original by Reinie Ern . MR 3155456



Ralph Abraham and Jerrold E. Marsden, *Foundations of mechanics*, second ed., Benjamin/Cummings Publishing Co., Inc., Advanced Book Program, Reading, MA, 1978, With the assistance of Tudor Ra iu and Richard Cushman. MR 515141

References ii

-  V. I. Arnold, *Mathematical methods of classical mechanics*, Graduate Texts in Mathematics, vol. 60, Springer-Verlag, New York, [1989?], Translated from the 1974 Russian original by K. Vogtmann and A. Weinstein, Corrected reprint of the second (1989) edition. MR 1345386
-  Ana Cannas da Silva, *Lectures on symplectic geometry*, Lecture Notes in Mathematics, vol. 1764, Springer-Verlag, Berlin, 2001. MR 1853077
-  A. Celletti, *Stability and chaos in celestial mechanics*, Springer Praxis Books, Springer Berlin Heidelberg, 2010.
-  Urs Frauenfelder and Otto van Koert, *The restricted three-body problem and holomorphic curves*, Pathways in Mathematics, Birkhäuser/Springer, Cham, 2018. MR 3837531

References iii

-  Myeonggi Kwon and Otto van Koert, *Brieskorn manifolds in contact topology*, Bull. Lond. Math. Soc. **48** (2016), no. 2, 173–241. MR 3483060
-  Dongho Lee, *Conley-zehnder indices of spatial rotating kepler problem*, 2025.
-  J. Milnor, *Morse theory*, Annals of Mathematics Studies, vol. No. 51, Princeton University Press, Princeton, NJ, 1963, Based on lecture notes by M. Spivak and R. Wells. MR 163331
-  S. Smale, *Differentiable dynamical systems*, Bull. Amer. Math. Soc. **73** (1967), 747–817. MR 228014